

Role of *Slc13a5* in osteoblast-mediated bone remodeling and marrow fat regulation

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Introduction: Bone remodeling is a dynamic and highly regulated process that enables bone to adapt to mechanical and metabolic stimuli. Mechanical loading is a primary driver of this process, stimulating bone formation. When this mechanoadaptive response is impaired, bone becomes more susceptible to deterioration, ultimately contributing to pathological conditions such as osteoporosis and increased fracture risk. Understanding the cellular and molecular mechanisms that govern load-induced bone remodeling is therefore essential for identifying factors that preserve skeletal health.

At the tissue level, bone strength is determined by its composition and geometry. Citrate is a critical component of the bone matrix and contributes to overall bone strength and integrity. In addition to its structural role, citrate is a key intermediate in cellular metabolism, linking energy production to metabolic processes as it plays a role in the Tricarboxylic Acid Cycle. Osteoblasts, the primary bone-forming cells, rely on tightly regulated metabolic pathways to support matrix production and mineralization.

Citrate uptake into cells is mediated by the sodium-dependent citrate transporter (NaCT) encoded by the *Slc13a5* gene. This transporter plays a central role in regulating intracellular citrate levels, thereby influencing metabolic activity. In osteoblasts, citrate transport may be particularly important for coordinating structural demands of bone formation. However, the role of osteoblast-specific citrate transport in regulating bone remodeling remains unclear.

In addition to their role in bone formation, osteoblasts are closely linked to adipocytes through a shared mesenchymal progenitor. This relationship suggests that shifts in osteoblast metabolism may influence lineage allocation, potentially altering the balance between bone formation and marrow adiposity. As a result, altered citrate transport in osteoblasts may have downstream effects on marrow fat accumulation.

Thus, it is unknown whether citrate plays a key role in regulating bone structure and marrow fat balance. Using a tibial loading model in combination with microcomputed tomography (microCT) and spatial analysis techniques, we evaluated both structural and metabolic outcomes in control mice (Cre⁻) that have the channel encoded by the *Slc13a5* gene present and conditional knockout (cKO) mice which do not have the channel present. By impairing citrate uptake in osteoblasts, we hypothesized that impaired citrate transport in osteoblasts would blunt the bone

remodeling response to mechanical loading and promote adipocyte formation within the bone marrow.

Within this project, I was responsible for analyzing *in vivo* and *ex vivo* microCT scans, image registration, and marrow adiposity quantification.

The combined use of *in vivo* and *ex vivo* microCT provides complementary advantages. *In vivo* imaging enables longitudinal tracking of bone remodeling within the same animal, allowing direct comparison of pre- and post-loading conditions. In contrast, *ex vivo* microCT offers higher resolution, enabling more precise quantification of structural parameters. Together, these approaches allow both dynamic and high-resolution assessment of bone adaptation.

Methods:

To first understand what we are quantifying, we must define our parameters.

Parameters:

Bone Mineral Density:

Bone mineral density (BMD) represents the concentration of mineralized tissue within a given volume of bone and serves as a key indicator of overall bone quality. BMD reflects the degree of mineralization of the bone matrix, which directly influences the stiffness and strength of the tissue. Higher BMD values are generally associated with increased resistance to deformation and fracture, while lower BMD indicates reduced mineral content and greater susceptibility to mechanical failure. Unlike purely geometric parameters, BMD captures material properties of bone, providing complementary insight into structural wholeness. Mechanical loading is known to promote increases in BMD, making it a valuable metric for assessing bone adaptation. Thus, changes in BMD reflect how mechanical loading and molecular factors influence the material composition and mechanical competence of bone.

Cortical Thickness:

Cortical thickness (Ct.Th) represents the average thickness of the cortical bone shell at a given cross-section along the tibia. Because the cortical compartment bears the majority of mechanical load in long bones, Ct.Th is a primary determinant of bending and compressive strength. Increases in cortical thickness generally reflect enhanced bone formation or reduced bone resorption, resulting in a greater capacity to withstand applied mechanical forces. Mechanical loading is known to stimulate cortical thickening as an adaptive response, making Ct.Th a sensitive metric for evaluating mechanoadaptation.

Imin:

The minimum second moment of area (Imin) represents the resistance of a bone cross-section to bending about its weakest principal axis. Imin is important for understanding structural robustness and directional symmetry of the bone.

Changes in Imin indicate whether adaptation occurs uniformly across multiple bending directions or is primarily localized along a dominant loading axis. Imin provides a more complete assessment of how treatments and mechanical loading influence bone geometry and mechanical competence.

Trabecular Thickness:

Trabecular thickness (Tb.Th) represents the average thickness of the individual trabeculae within the cancellous (spongy) bone compartment. Unlike cortical bone, trabecular bone forms a porous, lattice-like network that is highly metabolically active and responsive to mechanical and physiological stimuli. Tb.Th is a key determinant of trabecular bone strength, as thicker trabeculae contribute to improved load distribution and resistance to microstructural failure. Increases in trabecular thickness generally reflect enhanced bone formation within the trabecular compartment or reduced resorption of existing trabeculae, resulting in a more robust internal bone architecture. Mechanical loading has been shown to promote trabecular thickening as part of the adaptive remodeling response, making Tb.Th an excellent indicator of changes in bone microstructure.

The selected structural parameters provide complementary insight into bone adaptation. Cortical thickness and trabecular thickness reflect changes in bone mass and architecture, while the second moment of area captures how bone material is distributed relative to bending forces. Bone mineral density, in contrast, reflects the material composition of bone tissue. Together, these metrics enable differentiation between structural adaptation driven by geometric changes and alterations in bone quality related to mineralization

Now that we've provided context for our project, we can discuss the different components that were used to perform this experiment.

Animal Model:

When performing these experiments, our lab chose to use mice as the experimental model because their bone remodeling process closely parallels those of humans, and they allow for precise control over systemic treatments and mechanical loading conditions. As opposed to cell culture systems, mice provide an intact physiological environment where hormonal, metabolic, and mechanical factors interact to regulate bone adaptation. Additionally, the ethics of gene alterations combined with the limitations in performing controlled mechanical loading experiments makes mice the best fit for our projects.

Osteoblast-specific conditional knockout mice lacking *Slc13a5* gene (cKO) were used to investigate the role of citrate transport in bone remodeling. Cre recombinase-negative (Cre⁻) littermate controls were used as a comparison group. This model allows for targeted disruption of citrate uptake specifically in osteoblasts, enabling isolation of its role in bone formation without systemic metabolic effects.

Tibial Loading Model:

To evaluate the bone remodeling response to mechanical stimuli, an *in vivo* tibial loading model was employed (Figure 1). For each mouse, the right tibia was subjected to cyclic mechanical compressions, while the left tibia served as a non-loaded internal control. Loading was performed five days per week over a two-week period. This approach enables direct within-animal comparison, reducing variability and allowing assessment of load-induced adaptation across genotypes.

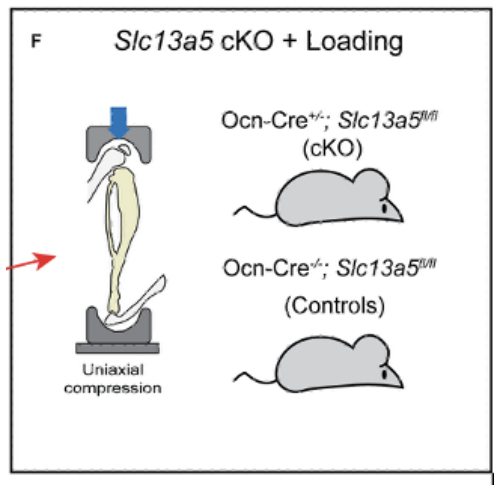


Figure 1: Schematic of the *in vivo* tibial loading model. The right tibia was subjected to cyclic uniaxial compression, while the left tibia served as a non-loaded internal control. Osteoblast-specific *Slc13a5* conditional knockout (cKO) mice and Cre⁻ controls were used to evaluate the effects of impaired citrate transport on load-induced bone remodeling. Adapted from Ref [1].

***In Vivo* microCT Imaging and Registration:**

To create bone scans, our lab utilized micro computed tomography (microCT) imaging to quantify structural outcomes and create comparisons among experimental groups. MicroCT provides high-resolution, three-dimensional visualization of bone architecture, allowing for precise quantification of cortical geometry.

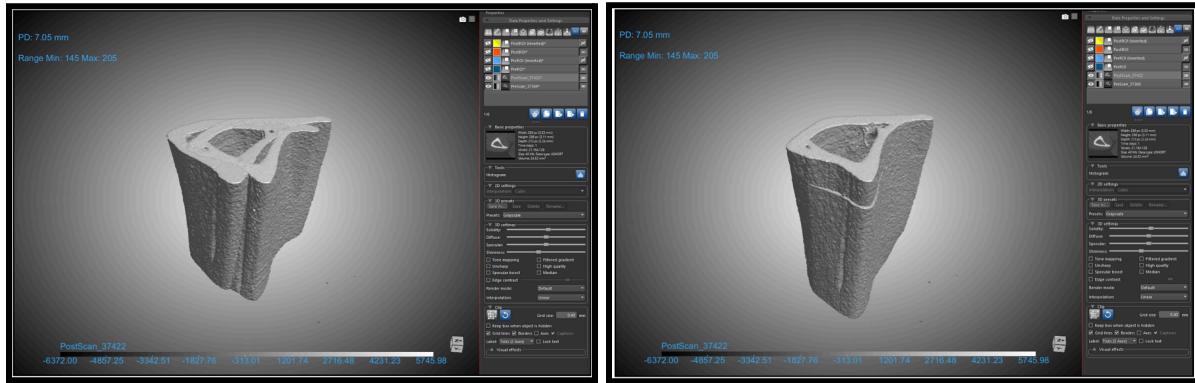


Figure 2: Representative *in vivo* microCT reconstructions of the tibia before (left) and after (right) image registration using Dragonfly software. Registration enables spatial alignment of sequential scans, allowing for identification and quantification of localized bone remodeling events.

In vivo microCT imaging was utilized to monitor bone remodeling over time. Sequential scans were aligned using Dragonfly software to enable precise spatial registration of bone structures. This process allows for identification of localized regions of bone formation and resorption by comparing changes between time points. The resulting spatial maps provide insight into the distribution of remodeling activity across the bone.

This process involved aligning the centroids between pre- and post-loading scans (Figure 2), cropping to the region just below the growth plate, and using computational features to quantify formation and resorption within the endosteal and periosteal surfaces, as illustrated in Figure 3.

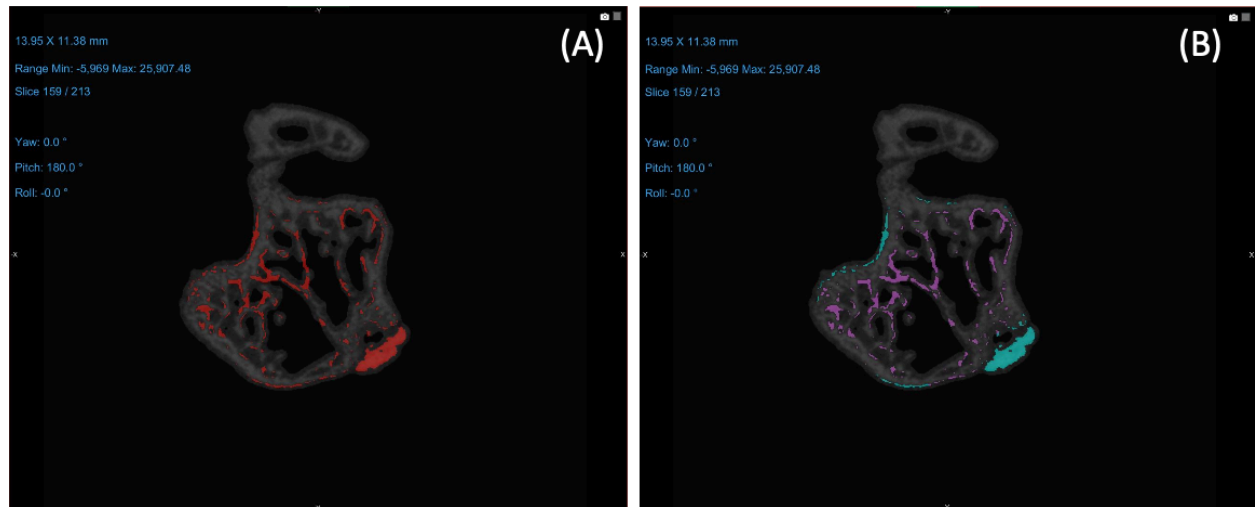


Figure 3: Representative spatial mapping of bone formation in a *Slc13a5* cKO mouse following mechanical loading. (A) Total bone formation is shown as a red overlay. (B) Spatial distribution of bone formation, with periosteal formation indicated in blue and endosteal formation indicated in magenta. This mapping enables differentiation of remodeling activity across distinct bone surfaces.

***Ex Vivo* microCT Analysis:**

Following completion of the loading protocol, *ex vivo* microCT scans were analyzed using ScanCo Evaluation software to quantify structural bone parameters. Measurements included bone mineral density, cortical thickness, trabecular thickness, and second moment of area. These parameters provide quantitative assessment of bone geometry and mechanical responsiveness, allowing comparison between loaded and non-loaded limbs as well as between genotypes.

Marrow Adiposity Quantification

Bone marrow adiposity was assessed using osmium tetroxide staining, which selectively labels lipid-rich adipose tissue on high contrast images within decalcified bone marrow. Stained samples were imaged and analyzed to quantify bone marrow adipose tissue volume using ScanCo Evaluation software. This method enables visualization and measurement of marrow fat content, providing insight into potential shifts in mesenchymal lineage allocation between osteoblasts and adipocytes.

To ensure consistent anatomical sampling, a standardized landmark corresponding to the proximal tibia-fibula junction was used to define the region of interest. Specifically, the slice at which the tibia and fibula cross-sections first appeared to contact was identified as a reference point. From this location, the region of interest was defined by moving 10 slices proximally to avoid variability associated with the growth plate and epiphyseal region. A total of 200

consecutive slices were then contoured and analyzed using a consistent threshold. This approach ensured consistent sampling of the metaphyseal marrow compartment across all specimens.

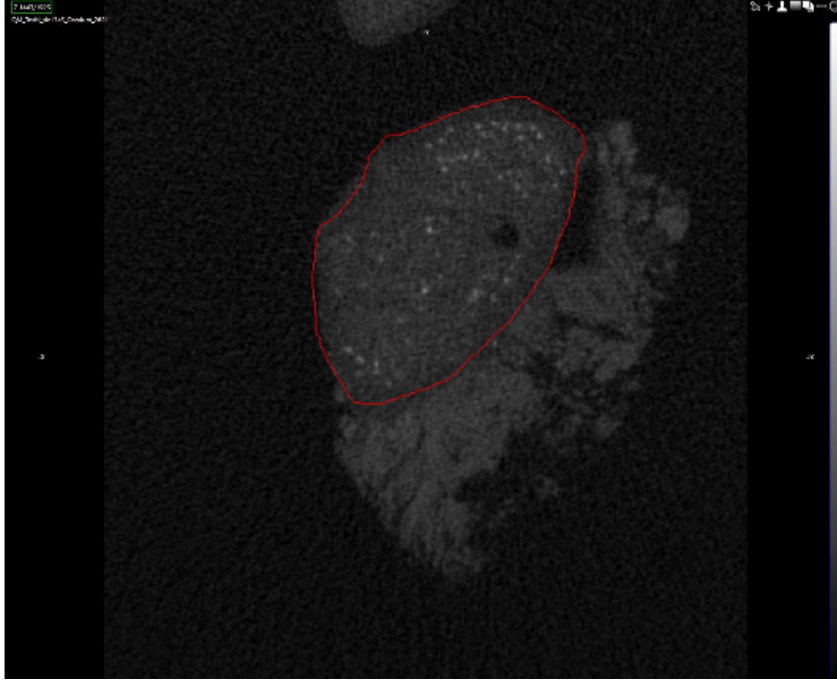


Figure 4: Representative osmium tetroxide-stained microCT image of the proximal tibia used for marrow adiposity quantification. The red outline indicates the contoured region of interest within the marrow cavity, where bone marrow adipose tissue (BMAT) volume was measured using a consistent threshold across samples.

Outcome Measurement Methodology

Bone remodeling activity was quantified using formation and resorption volume percentages, normalized to pre-existing bone volume (F/Prebone and R/Prebone). These metrics capture dynamic changes in bone remodeling by measuring the relative contributions of bone formation and resorption. Together with structural parameters and BMAT measurements, these outcomes provide a comprehensive assessment of both the mechanical and metabolic effects of osteoblast-specific *Slc13a5* deletion.

Reproducibility and Data Consistency Considerations:

To ensure consistency across analyses, all image processing and quantification steps were performed using standardized parameters and thresholds across samples. Registration, contouring, and region-of-interest selection were conducted using the same anatomical landmarks and procedural workflow to minimize user-dependent variability. In particular, alignment of pre- and post-loading scans was carefully verified to ensure accurate spatial comparison of remodeling events.

Additionally, normalization of remodeling metrics to pre-existing bone volume (F/Prebone and R/Prebone) was used to account for baseline differences between specimens. This approach allows for more meaningful comparison of relative changes in remodeling activity across genotypes and loading conditions. By maintaining consistent analysis parameters and normalization strategies, variability introduced by technical factors was minimized, allowing observed differences to more accurately reflect biological effects.

Results & Discussion:

Structural Response to Mechanical Loading:

Ex vivo microCT analysis revealed that mechanical loading induced increases in structural bone parameters in both *Slc13a5* cKO and Cre⁻ control mice, as illustrated in Figure 5. Specifically, cortical thickness, trabecular thickness, and the minimum second moment of area values were all higher in loaded legs compared to their respective non-loaded controls. These increases were observed across both genotypes, indicating a consistent adaptive response to mechanical stimuli.

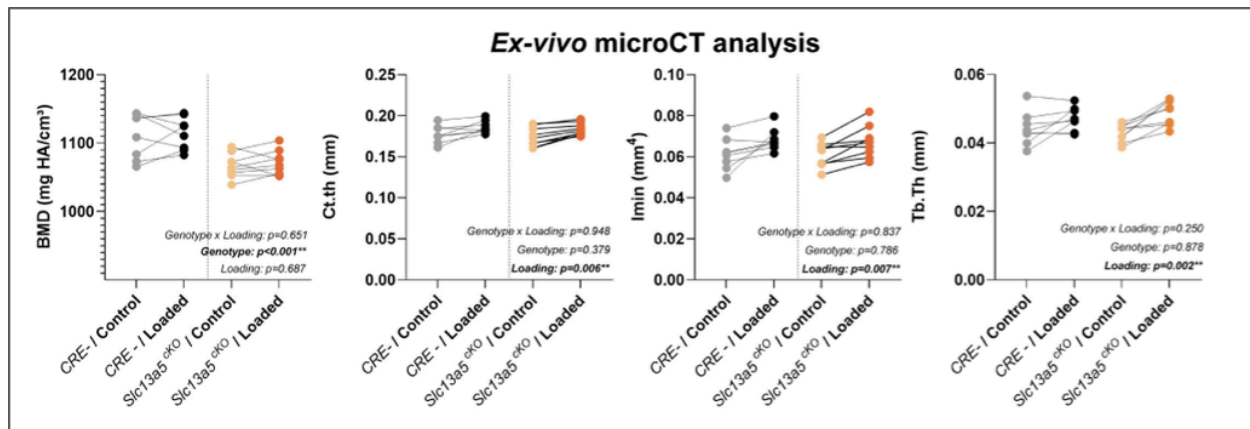


Figure 5: *Ex vivo* microCT analysis of structural bone parameters in Cre⁻ control and *Slc13a5* cKO mice under loaded and non-loaded conditions. Measurements include bone mineral density, cortical thickness, minimum second moment of area, and trabecular thickness. Mechanical loading resulted in increases in Ct.Th, Imin, and Tb.Th in both genotypes, with no significant genotype × loading interaction observed. A reduction in BMD was observed in cKO mice relative to controls. Adapted from Ref. [1].

Despite these load-induced changes, no significant differences were observed between cKO and control mice in their response to mechanical loading. Both groups exhibited comparable increases in structural parameters, suggesting that osteoblast-specific deletion of *Slc13a5* does not impair the bone's ability to adapt to mechanical stimuli. However, a reduction in bone mineral density was observed in cKO mice relative to controls, indicating that while structural adaptation is preserved, material properties of the bone may be altered.

These findings suggest that citrate transport in osteoblasts is not essential for load-induced structural adaptation of bone. The preservation of increases in Ct.Th, Tb.Th, and Imin in cKO mice indicates that pathways driving bone formation remain functional despite impaired citrate uptake. However, the observed decrease in BMD may reflect subtle alterations in matrix mineralization or bone quality that are not captured by geometric parameters alone.

Total Bone Remodeling Activity

As shown in Figure 6, both formation and resorption were increased in response to mechanical loading in both *Slc13a5* cKO and Cre⁻ control mice. Specifically, formation volume was normalized to pre-existing bone (F/Prebone) as well as resorption volume (R/Prebone) across both genotypes.

Despite these load-induced increases, no significant differences were concluded between cKO and control groups for either F/Prebone or R/Prebone. These findings indicate that the overall magnitude of bone remodeling activity is preserved following osteoblast-specific deletion of *Slc13a5*.

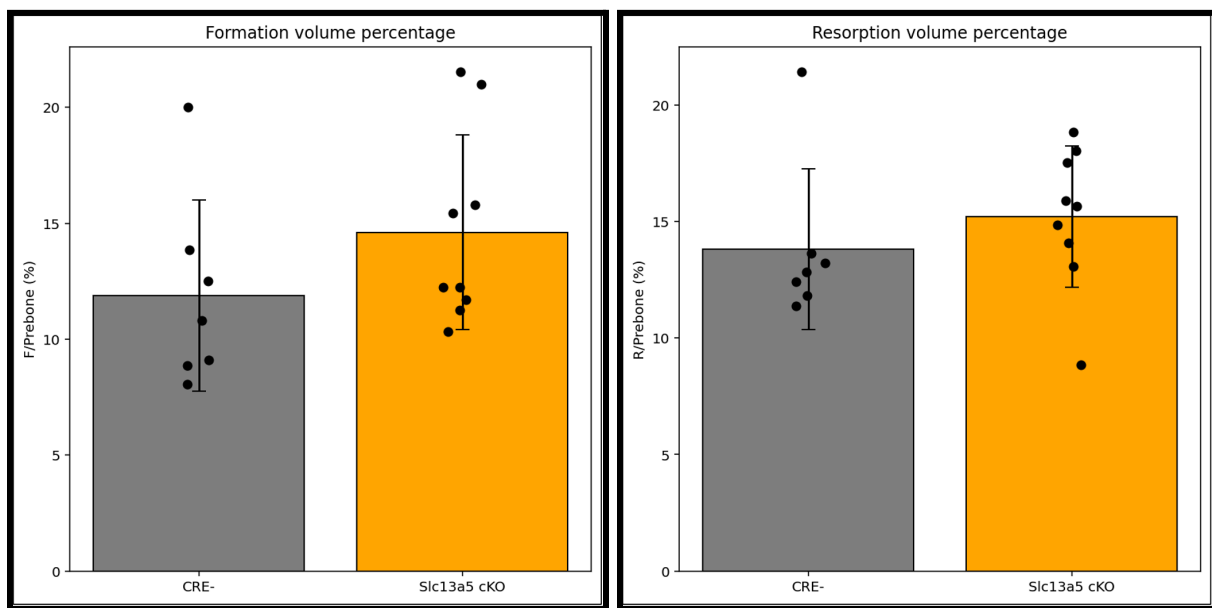


Figure 6: Quantification of total bone remodeling activity in Cre⁻ control and *Slc13a5* cKO mice. (Left) Formation volume normalized to pre-existing bone (F/Prebone). (Right) Resorption volume normalized to pre-existing bone (R/Prebone). Individual data points are shown with bars representing mean $\pm$ 1 standard deviation. No significant differences was concluded between genotypes.

The preservation of both formation and resorption activity suggests that citrate transport in osteoblasts is not required for the initiation or execution of load-induced remodeling. These results are consistent with the structural findings, further supporting the conclusion that mechanoadaptive responses remain intact despite disruption of citrate uptake.

These results also provide insight into the balance between bone formation and resorption during mechanoadaptation. Under normal physiological conditions, bone remodeling is tightly regulated

such that formation and resorption are coupled processes. The simultaneous increase in both F/Prebone and R/Prebone observed in this study suggests that mechanical loading stimulates an overall increase in remodeling rather than selectively enhancing formation alone. This coordinated response is essential for maintaining bone structure.

The preservation of this balance in Slc13a5 cKO mice further suggests that other upstream signaling pathways responsible for initiating remodeling remain functional despite metabolic disruption. It is possible that compensatory metabolic pathways within osteoblasts support bone formation in the absence of normal citrate uptake, thereby maintaining overall remodeling activity.

Spatial Distribution of Remodeling Activity

To further characterize remodeling dynamics, formation and resorption were analyzed separately at the endosteal and periosteal surfaces. Spatial mapping and quantitative analysis revealed that bone formation occurred at both surfaces in response to mechanical loading (Figure 7).

Endosteal and periosteal formation volumes (EndoF/Prebone and PerioF/Prebone) exhibited similar trends between cKO and control mice, with no significant differences observed between genotypes.

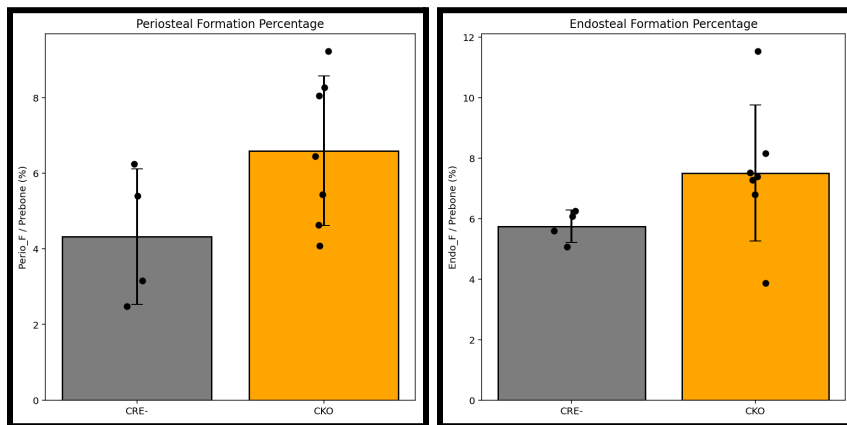


Figure 7: Quantification of surface-specific bone formation in Cre⁻ control and *Slc13a5* cKO mice. (Left) Periosteal formation volume normalized to pre-existing bone (PerioF/Prebone). (Right) Endosteal formation volume normalized to pre-existing bone (EndoF/Prebone). Individual data points are shown with bars representing mean $\pm$ 1 standard deviation. No significant difference was concluded between genotypes at either surface.

Likewise, resorption activity at both surfaces (EndoR/Prebone and PerioR/Prebone) remained comparable across groups.

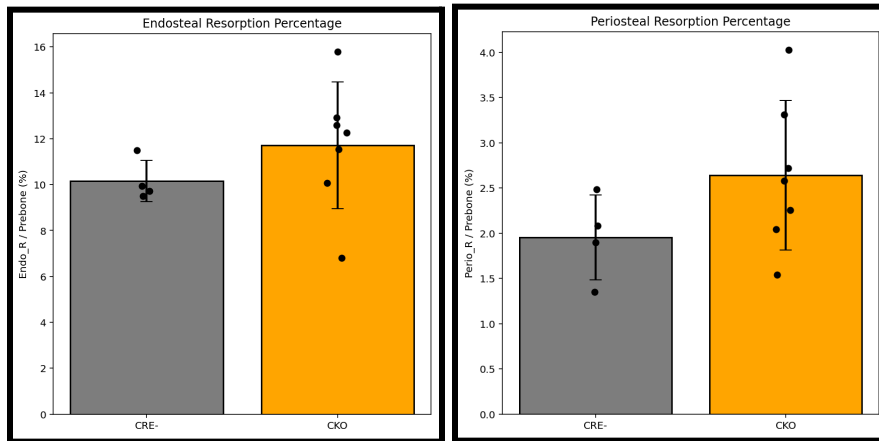


Figure 8: Quantification of surface-specific bone resorption in Cre⁻ control and *Slc13a5* cKO mice. (Left) Endosteal resorption volume normalized to pre-existing bone (EndoR/Prebone). (Right) Periosteal resorption volume normalized to pre-existing bone (PerioR/Prebone). Individual data points are shown with bars representing mean $\pm$ 1 standard deviation. No significant differences was concluded between genotypes at either surface.

These findings indicate that osteoblast-specific deletion of *Slc13a5* does not alter the spatial distribution of bone remodeling activity, and that citrate transport does not play a major role in directing where remodeling occurs within the bone.

Marrow Adiposity

After bone marrow adipose tissue volume was quantified using osmium staining, no significant differences in BMAT volume were observed between *Slc13a5* cKO and control mice.

This indicates that osteoblast-specific deletion of *Slc13a5* does not significantly influence marrow adiposity under the conditions tested. Despite a slight trend toward increased BMAT volume in cKO mice, this difference did not reach statistical significance, suggesting that impaired citrate transport in osteoblasts does not drive a shift toward adipocyte formation within the bone marrow.

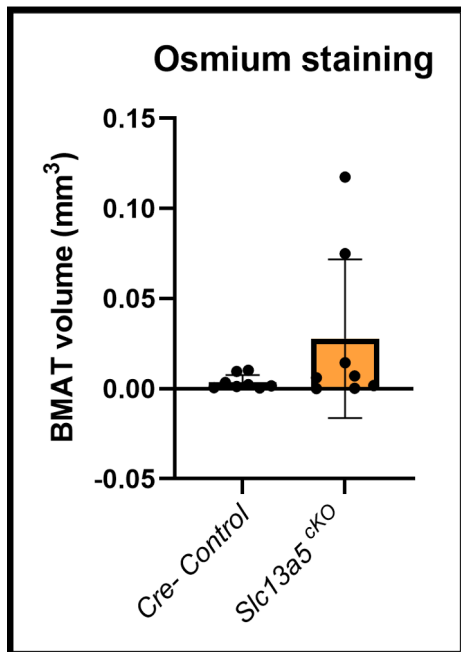


Figure 9: Quantification of bone marrow adipose tissue volume in Cre⁻ control and *Slc13a5* cKO mice using osmium tetroxide staining. Individual data points are shown with bars representing mean $\pm$ 1 standard deviation. No significant differences in BMAT volume were observed between genotypes, although a slight increase was observed in cKO mice.

Conclusion & Future Steps:

We initially hypothesized that impaired citrate transport in osteoblasts would blunt the bone remodeling response to mechanical stimuli and promote adipocyte formation within the bone marrow.

Contrary to this hypothesis, our findings demonstrate that osteoblast-specific deletion of *Slc13a5* does not impair the structural or dynamic response of bone to mechanical loading. *Ex vivo* microCT analysis revealed that both cKO and control mice exhibited comparable increases in cortical thickness, trabecular thickness, and bending resistance parameters following loading. Similarly, total remodeling activity, as measured by F/Prebone and R/Prebone, was preserved across genotypes, indicating that the magnitude of bone remodeling remains intact despite disruption of citrate transport.

Further spatial analysis from *In Vivo* microCT analysis showed that bone formation and resorption occurred at both endosteal and periosteal surfaces in both groups, with no significant differences in surface-specific remodeling activity. These findings suggest that *Slc13a5* does not play a major role in directing the localization of bone remodeling. Additionally, marrow adiposity analysis revealed no significant differences in BMAT volume between cKO and control mice, indicating that osteoblast-specific citrate transport does not significantly influence mesenchymal lineage allocation toward adipogenesis under the conditions tested.

Notably, a reduction in bone mineral density was observed in cKO mice, suggesting that while structural adaptation and remodeling dynamics are preserved, citrate transport may contribute to aspects of bone material composition or matrix mineralization.

Together, these findings indicate that osteoblast-specific citrate transport is not required for load-induced bone remodeling or regulation of marrow adiposity, but may play a more subtle role in maintaining bone material properties. Future studies should further investigate the mechanistic link between citrate metabolism and bone matrix composition, potentially through analysis of mineralization patterns. Also, future studies could investigate how citrate transport influences osteoblast metabolism under varying physiological conditions, such as aging, metabolic stress, or disease states. Expanding sample size and incorporating additional molecular or histological analyses may also help uncover subtle genotype-dependent effects that were not detected in this study.

These findings also highlight an important distinction between bone quantity and bone quality. While geometric and structural adaptations to loading were preserved, the observed reduction in BMD suggests that citrate transport may contribute to the material properties of bone, such as mineralization or matrix composition. This distinction is clinically relevant, as changes in bone quality can influence fracture risk independently of structural parameters.

Additionally, these results emphasize the complexity of the relationship between metabolism and skeletal adaptation. While metabolic pathways are essential for cellular function, their role in regulating large-scale physiological processes such as mechanoadaptation may be more nuanced than we initially expected.

References:

[1] Meslier, Q. A., et al. (2026). *Spatial transcriptomics for gene discovery identifies Slc13a5 as a modulator of bone mechanoadaptation*. bioRxiv. <https://doi.org/10.64898/2026.03.11.711126>

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I also want to thank Dr. Erica Scheller for giving me the opportunity to work in her lab and for creating such a positive and welcoming environment. Being part of the lab has been a great experience, and I've really appreciated the chance to contribute while learning more about how research is actually carried out.

Through this independent study, I gained hands-on experience in quantitative bone imaging, data normalization, and structural analysis. I became more comfortable working with microCT data, selecting relevant parameters, and interpreting biomechanical metrics. I also improved my ability to communicate results through figure creation and presenting at lab meetings with Dr. Scheller. Overall, this experience helped me grow both technically and in how I think about research.

This project helped me better understand how different factors come together to influence bone adaptation, especially how changes at the cellular level can affect overall structure and function. Continuing to be involved in projects like these, especially with data analysis, will help prepare me as I work towards becoming a PhD level researcher.